

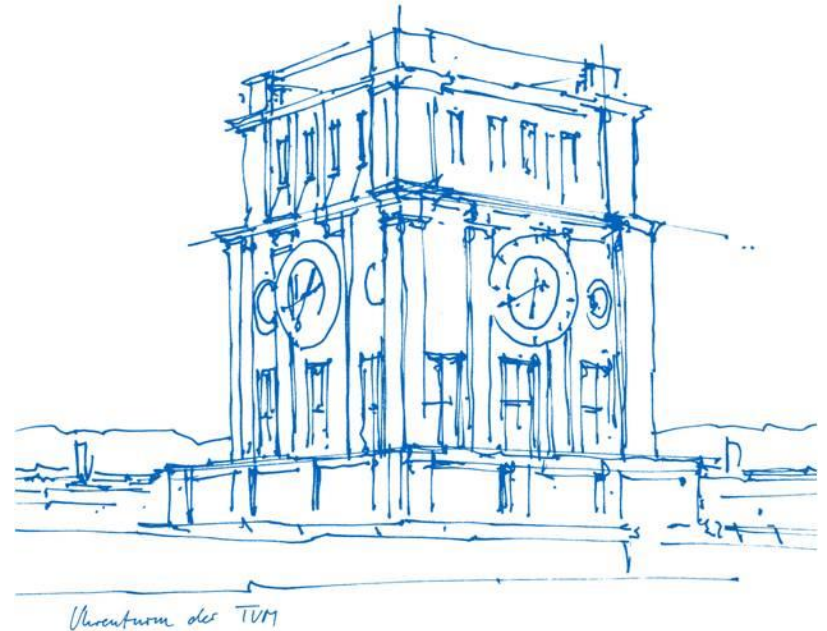
Advanced Practical Course: Development of AI and Data-driven Applications at the Example of ERPsim

Organization Slides – Summer Term 2026

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TUM School of Computation, Information and Technology
Technical University of Munich

April 15th , 2026



Agenda

1 **Instructors and Context**

2 **Course Content**

3 **Course Structure**

4 **Grading**

5 **Course Material**

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Your Instructors



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The SAP UCC Munich

The SAP UCC Munich is an **Education-as-a-Service Provider** in the area of SAP

- Operate and administrate more than **100 SAP systems** for **350 universities** on our own hardware
- Develop curricula to teach students various concepts on the example of SAP software e.g.
 - Enterprise Resource Planning processes (Controlling, Sales, Production, ...)
 - Enterprise Software Engineering
 - AI in Enterprise Software
 - Cross System Integration
 - and many more ...



It's time to get to know you!

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Three Content Pillars of the Course

Enterprise Systems and Integrated Business Processes

- Understand the **integration of operational business processes and accounting processes**.
- Understand **how integrated enterprise systems work**.
- Gain **hands-on experience** with an enterprise system (i.e., SAP S/4HANA).

Data Architecture

- Understand the **role of data architecture in a company**.
- Understand **data architecture paradigms**.
- Understand the **concept of data products**.

Data Analytics and Machine Learning

- Understand the **fundamentals of machine learning and genAI**.
- Understand how **analytical applications can be used for better decision-making** in organizations.
- Gain experience in **designing and developing such an application**.

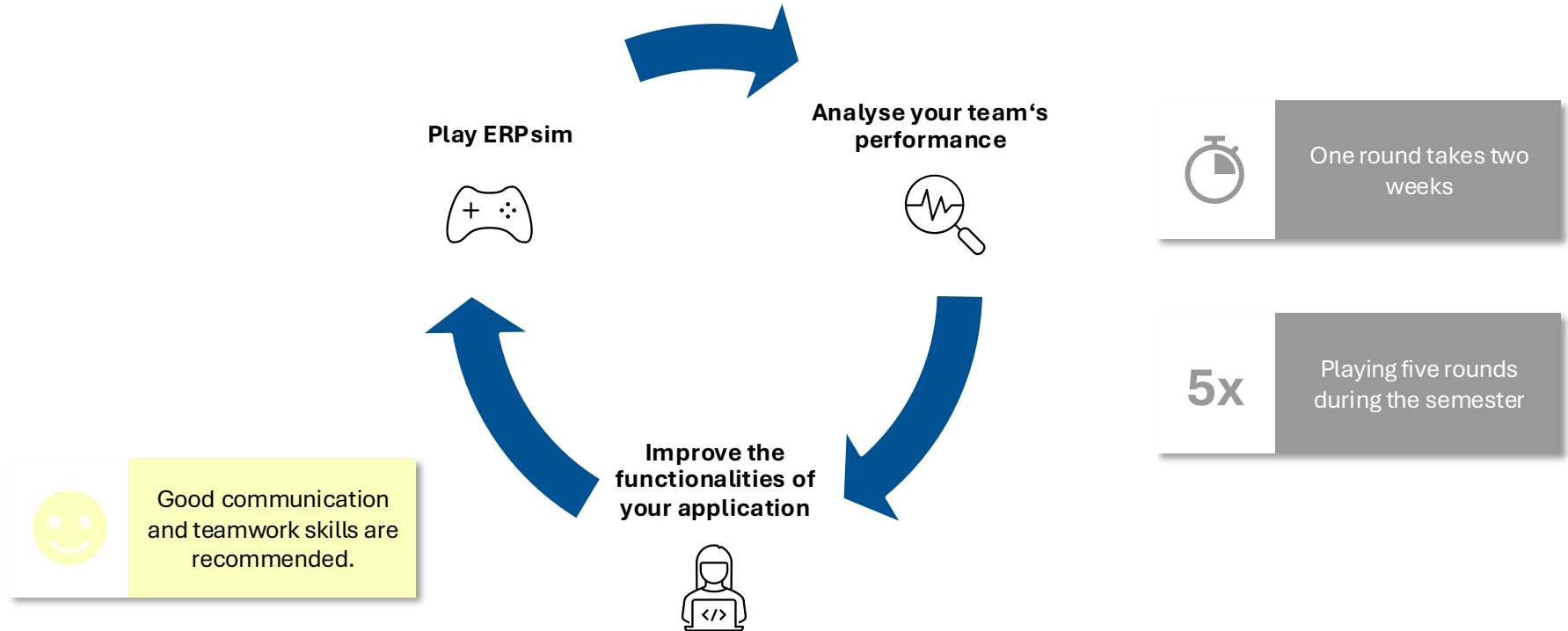
Heart of the Course: ERPsim

- ERPsim is a business simulation game running on a real SAP S/4HANA system.
- Students manage a virtual company and make **data-driven operational decisions**.
- Teams compete over several rounds to improve the **performance and profitability** of their company.



Course Objective

→ Develop an analytical application to improve your team's performance in ERPsim.



Exemplary Technological Concepts to Use

Programming Languages / Application Runtime

- Java
- Python
- Node.js

Enterprise Application Frameworks

- SAP Cloud Application Programming Model

Business Transformation Tools

- SAP Signavio
- SAP LeanIX

AI and Machine Learning Platform

- IBM watsonx

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Building Blocks of this Practical Course

Introduction Weeks

15.04.2026 – 21.04.2026

Get in touch with enterprise systems and business processes, as well as data analytics and machine learning fundamentals.

Knowledge Check

04.05.2026

An individual multiple-choice test regarding the lectures during the introduction weeks.

Project Work

27.04.2026 – 20.07.2026

Development of an application to improve your team's performance in ERPsim.

Part 1: Introduction Weeks Appointments

Introduction Weeks

15.04.2026 – 21.04.2026

Get in touch with enterprise systems and business processes, as well as data analytics and machine learning fundamentals.

The introductory weeks are structured in block lectures

Tuesday	15.04.2026	09:00 – 12:15	FMI 01.13.010	Intro to Business Processes
Thursday	16.04.2026	09:00 – 13:00	BC2 0.01.16	Intro to CAP
Friday	17.04.2026	09:00 – 13:00	BC2 0.01.16	Intro to Data Architecture
Monday	20.04.2026	09:00 – 13:00	FMI 01.13.010	Intro to watsonx (ML)
Monday	20.04.2026	13:00 – 17:00	SAP Labs	SAP Experience Center
Tuesday	21.04.2026	09:00 – 11:30	FMI 03.09.012	Intro to watsonx (genAI)

Further Information

- The introductory weeks consist of **guest lectures** and **lectures led by us**
- All lectures consist of a theory part and an exercise part



Sessions are important because we expect you to apply the concepts in your application.

Part 2: Knowledge Check

Knowledge Check

04.05.2026

An individual multiple-choice test regarding the lectures during the introduction weeks.

Important Dates for the Knowledge Check

Monday

04.05.2026

Knowledge Check

Further Information

- Individual multiple-choice test in Moodle
- Questions about the lectures during the introduction weeks
- 17 minutes for 15 questions
- Makes up 10% of your final grade
- Further information is shared during the course

Part 3: Project Work

Project Work

27.04.2026 – 20.07.2026

Development of an application to improve your team's performance in ERPsim.

Important Dates for the project work

Project Introduction Sessions

Monday	27.04.2026	09:00 – 12:00	FMI 01.13.007
Tuesday	28.04.2026	09:00 – 12:00	FMI 03.09.012

Concept Presentation

Monday	04.05.2026	15:00– 17:00	SAP Labs
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Agile Workshop

Thursday	07.05.2026	09:00 – 17:00	tba.
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ERPsim playing Sessions

Monday	11.05.2026	09:00 – 12:00	FMI 01.13.010
Thursday	28.05.2026	09:00 – 12:00	FMI 01.13.010
Tuesday	09.06.2026	09:00 – 12:00	FMI 01.09.014
Wednesday	24.06.2026	09:00 – 12:00	FMI 01.13.010
Tuesday	07.07.2026	09:00 – 12:00	tba.

Project Submission Deadline

Sunday	19.07.2026	23:59	
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Final Presentation

Monday	20.07.2026	09:00 – 12:00	SAP Labs
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Part 3: Project Work

Project Work

27.04.2026 – 20.07.2026

Development of an application to improve your team's performance in ERPsim.

Information about the project

- Teamwork in teams of 4 students
- Development of an application to improve your team's performance in ERPsim
- Presentation of your results
 - Bi-weekly status presentation within the paying sessions
 - Concept Presentation
 - Final Presentation
- Compilation of a project documentation
 - max. 15 pages using the template
 - One document per team
 - Format: PDF
 - Submission via Moodle



It is expected to apply the concept presented during the introduction weeks in your application.

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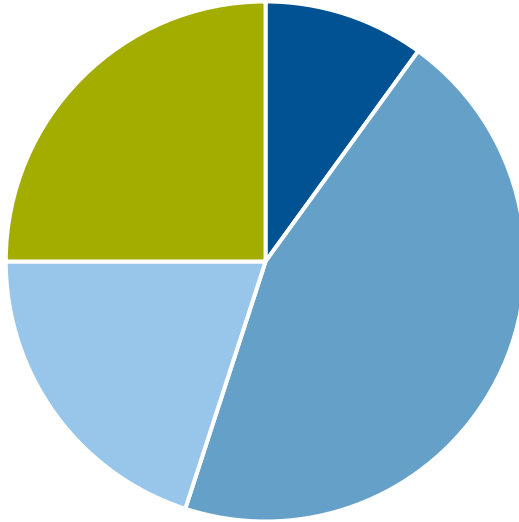
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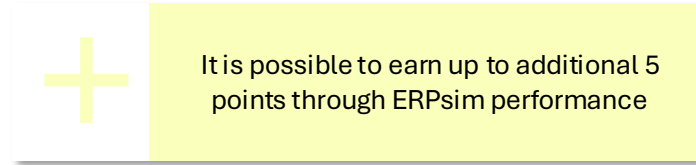
Grading Structure



It is possible to earn up to additional 5 points through ERPsim performance

- Overall: 100 points
- **Knowledge Check** (10 points)
- **Project work implementation** (45 points)
- **Project work documentation** (20 points)
- **Project presentation and discussion** (25 points)
- Participation of all team members required
 - Bi-weekly status presentation (5 points)
 - Concept presentation (5 points)
 - Final presentation (10 points)
 - Concept and final presentation discussion (5 points)
- Participation in all lecture events
 - absence **only permitted with preceding agreement** (e.g., by email)

Grading Bonus



5 points	Team with the overall highest ERPsim performance score
3 points	Team with the overall 2nd-highest ERPsim performance score
1 points	Team with the overall 3rd-highest ERPsim performance score
0 points	Team with the overall 4th-highest ERPsim performance score

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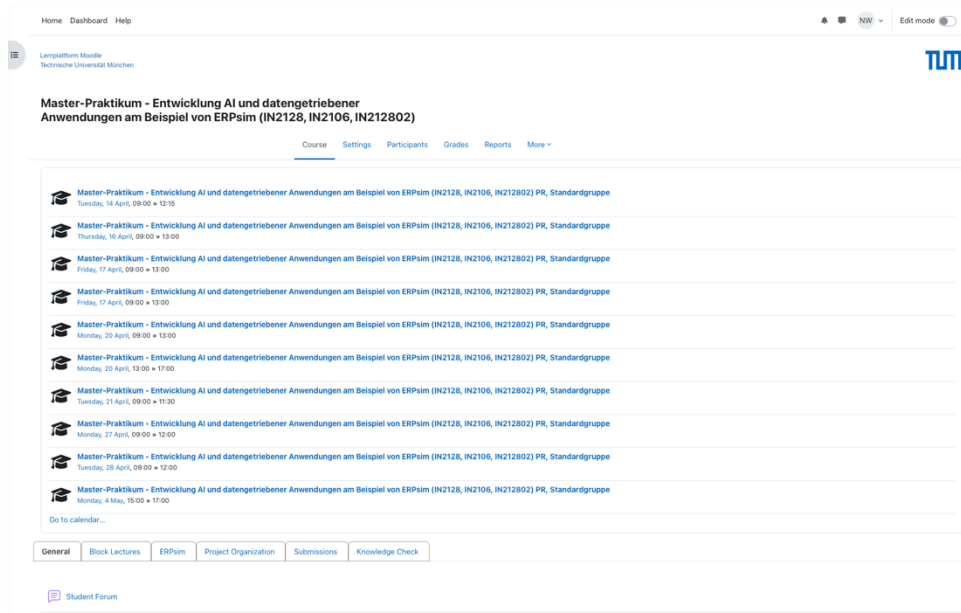
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www.moodle.tum.de



The screenshot shows a Moodle course page. At the top, there's a navigation bar with 'Home', 'Dashboard', and 'Help'. Below it, the course title is 'Master-Praktikum - Entwicklung AI und datengetriebener Anwendungen am Beispiel von ERPsm (IN2128, IN2106, IN212802)'. The course is part of the 'Langzeitstudien' program at 'Technische Universität München'. The course page has tabs for 'Course', 'Settings', 'Participants', 'Grades', 'Reports', and 'More'. The main content area lists several 'Master-Praktikum' sessions with their dates and times. At the bottom, there are tabs for 'General', 'Block Lectures', 'ERPsm', 'Project Organization', 'Submissions', and 'Knowledge Check'. A 'Student Forum' link is also visible at the bottom left.

Use the **Moodle Discussion Forum** if you have problems or questions

=> [Help each other!](#)

Only ask us in case of technical issues
(e.g., system not available)

=> sap-analytics-i17@tum.de



Please note: Your university email address will be forwarded to

- SAP to enable system access for SAP LeanIX,
- ERPsim Lab Montreal for game participation, and
- IBM to access IBM watsonx platform.

You also need a **Google account** to participate in a prompt game.

Thank you!

Do you have questions?